

Brainstorming Topics

- **Air Pollution and Health Outcomes**

- Possible Question:
 - Can county-level air quality predict asthma, COPD, or poor health rates
- Dataset Idea:
 - EPA county-level AQI data and CDC PLACES health data
- Hypothesis (maybe):
 - Counties with worse air quality will have higher rates of asthma, COPD, or poor physical health
- Models to compare (maybe):
 - Linear regression predicting asthma/COPD rate from AQI
 - Multiple regression adding poverty, income, or education
 - Classification (also maybe lol): high vs low asthma county
- Strength:
 - It connects climate/environment + health and the datasets are public and county level (so lots of data). CDC places includes health outcomes and social determinants, while EPA AirData has AQI data
- Sources:
 - <https://www.cdc.gov/places/tools/data-portal.html>
 - https://aqs.epa.gov/aqsweb/documents/aqi/aqi_data.html

- **Socioeconomic Predictors of Health**

- Possible Question:
 - Can income, poverty, education, and unemployment predict county-level health outcomes?
- Dataset Idea:
 - Use CDC PLACES, which already includes health outcomes and some social/economic variables
- Hypothesis:
 - Counties with higher poverty and lower education levels will have worse outcomes
- Models to compare:
 - Linear regression
 - Ridge or Lasso regression
 - KNN regression or classification
- Strengths:
 - Can possibly knock this down only needing one dataset, it also directly fits the class methods, and is also very practical.

Research Question:

How do air quality metrics and weather conditions impact human health outcomes?

Project abstract (2-4 sentences):

This dataset contains 5,811 records covering information on the relationship between air quality, weather factors, and public health. This includes variables such as the Air Quality Index (AQI), concentrations of various pollutants, weather conditions, and health impact indicators. The main outcome variables are **HealthImpactScore** and **HealthImpactClass**, which describe the level of health impact based on air quality and other relevant factors.

Dataset:

<https://www.kaggle.com/datasets/rabieelkharoua/air-quality-and-health-impact-dataset>

This dataset contains the following numerical columns:

Air Quality Metrics

- **AQI**: Air Quality Index, a measure of how polluted the air currently is or how polluted it is forecast to become.
- **PM10**: Concentration of particulate matter less than 10 micrometers in diameter ($\mu\text{g}/\text{m}^3$).
- **PM2_5**: Concentration of particulate matter less than 2.5 micrometers in diameter ($\mu\text{g}/\text{m}^3$).
- **NO2**: Concentration of nitrogen dioxide (ppb).
- **SO2**: Concentration of sulfur dioxide (ppb).
- **O3**: Concentration of ozone (ppb).

Weather Conditions

- **Temperature**: Temperature in degrees Celsius ($^{\circ}\text{C}$).
- **Humidity**: Humidity percentage (%).
- **WindSpeed**: Wind speed in meters per second (m/s).

Health Impact Metrics

- **RespiratoryCases**: Number of respiratory cases reported.
- **CardiovascularCases**: Number of cardiovascular cases reported.
- **HospitalAdmissions**: Number of hospital admissions reported.

Target Variable: Health Impact Class

- **HealthImpactScore**: A score indicating the overall health impact based on air quality and other related factors, ranging from 0 to 100.
- **HealthImpactClass**: Classification of the health impact based on the health impact score:
 - 0: 'Very High' (HealthImpactScore $\geq$ 80)
 - 1: 'High' (60 $\leq$ HealthImpactScore < 80)
 - 2: 'Moderate' (40 $\leq$ HealthImpactScore < 60)
 - 3: 'Low' (20 $\leq$ HealthImpactScore < 40)
 - 4: 'Very Low' (HealthImpactScore < 20)

The dataset contains **continuous columns**: pollutant measurements (AQI, PM2.5, PM10, NO₂, SO₂, O₃), weather variables (temperature, humidity, wind speed), and health-related outcomes (respiratory cases, cardiovascular cases, hospital admissions), and a **categorical outcome column**: **HealthImpactClass**.

Hypotheses and Initial Expectations

Our main expectation is that worse air quality will be associated with worse public health outcomes. We expect higher AQI values and higher concentrations of pollutants such as PM2.5, PM10, NO₂, SO₂, O₃ to be associated with higher **HealthImpactScore** values. We also expect weather variables such as temperature, humidity, and wind speed to contribute to health outcomes, though we expect them to be weaker predictors than direct air pollution measurements.

Null Hypothesis

The null hypothesis is that air quality metrics and weather conditions have no meaningful relationship with health impact outcomes. By this we mean, AQI, pollutant concentrations, temperature, humidity, and wind speed have no significant relationship with health impact outcomes.

Alternative Hypothesis

The alternative hypothesis is that AQI, pollutant concentrations, temperature, humidity, and wind speed do have a significant relationship with health impact outcomes. More specifically, we expect higher AQI, PM2.5, and PM10 levels to be associated with higher HealthImpactScore values. We also expect pollutant variables such as NO₂, SO₂, O₃ to help explain variation in health outcomes. Weather variables may improve prediction when included with air quality metrics, but they are expected to be less influential than the direct air pollution variables.

Because the dataset has both **continuous targets** and a **categorical target**, we will use **Logistic Regression**, **Multiple Linear Regression**, **PCA**, and **Polynomial Regression**.

Cross-Validation:

We will use k-fold cross-validation to evaluate and compare the performance of different models. For each model, we will use standard 10-fold cross-validation. First, the observations in the dataset will be randomly shuffled to eliminate any ordering effects that may exist in the original dataset. After shuffling, the dataset will be divided into 10 approximately equal-sized folds. One fold is held out as the testing set, and the remaining 9 folds are combined to form the training set. The model is trained on the training set, and is evaluated on the testing fold. This process is repeated 10 times so that each fold serves as the testing set exactly once. At the end of the process, the prediction errors from all 10 iterations are averaged to estimate the overall model performance.

Logistic Regression:

- The target variable is categorical (HealthImpactClass).

Multiple Linear Regression:

- Predicting HealthImpactClass, RespiratoryCases, and HospitalAdmissions.

Polynomial Regression:

- See if the model is non-linear.

PCA:

- The dataset has too many columns, so we use PCA to simplify it.

Model Selection

Logistic Regression & Multiple Linear Regression:

- We decided to use lasso regularization for both logistic and multiple Linear regression, as we explained in the hypothesis that some features are expected to be less influential with health impact, and we might consider dropping them.
- We will compare/contrast different regularization strengths.

Polynomial Regression:

- we decide to compare between degrees 1(linear), 2, 3, 5 to evaluate if the relationship is non-linear.

PC Regression:

- we will compare between the top 1, 2, 3, 5, 10 PCs.

For all of the hyperparameters explained above, we will use cross-validation to select the best parameter that provides the lowest test set.